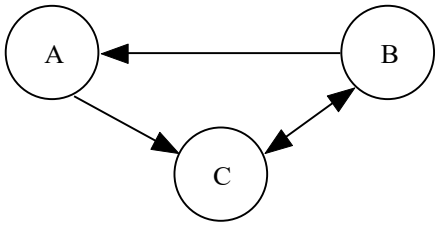
Compiled Problems –Senior Division

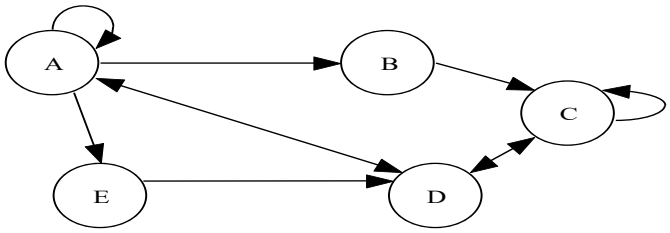
1. **00-01 C3 Graph Theory**

   How many paths of length 4 exist in the following graph?



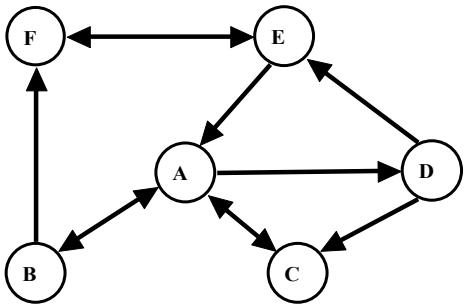
2. **00-01 C3 Graph Theory**

   Construct the adjacency matrix for the graph below



3. **01-02 C3 Graph Theory**

   How many 1's are in the adjacency matrix of the below graph?



4. **01-02 C3 Graph Theory**

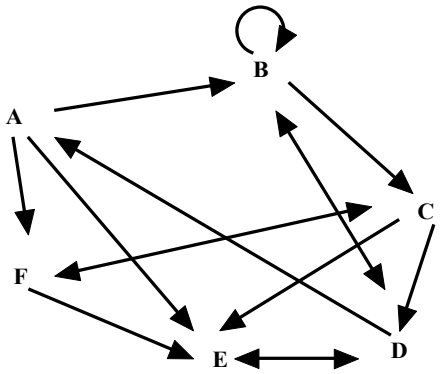
   A complete, undirected graph is one in which there is an undirected edge between every pair of distinct vertices. How many edges exist in a complete, undirected graph with 10 vertices?

5. **02-03 C3 Graph Theory**

   A complete, undirected graph is one in which there is an undirected edge between every pair of distinct vertices. How many edges are there in a complete undirected graph with 9 vertices?

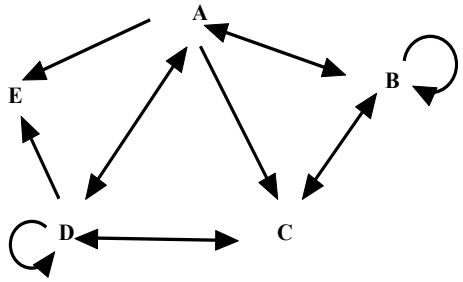
6. **02-03 C3 Graph Theory**

How many different cycles exist in the following directed graph?



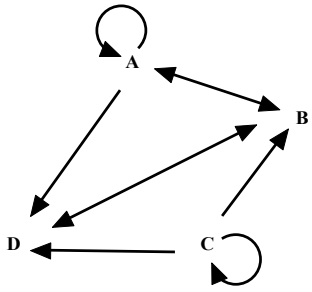
7. **03-04 C3 Graph Theory**

How many cycles are there in the following directed graph?



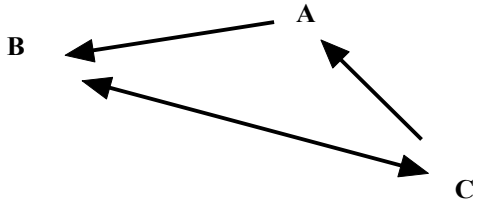
8. **03-04 C3 Graph Theory**

How many paths of length 3 exist in the following directed graph?



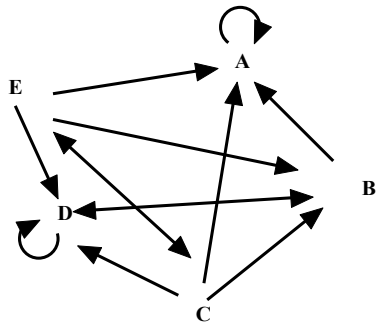
9. **04-05 C3 Graph Theory**

If the path AC is added, how many additional paths of length 2 will be added?



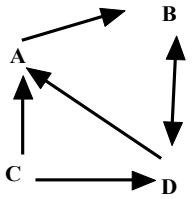
10. **04-05 C3 Graph Theory**

Counting the starting town, a salesperson visits 5 towns on his/her route.
After the 5th visit, the salesperson stays the night in that town, C.
List all of the possible starting points for the route.



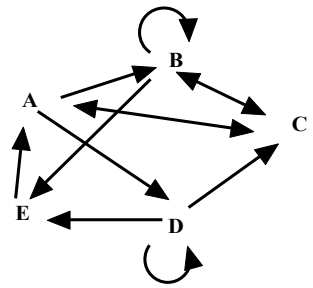
11. **05-06 C3 Graph Theory**

Given the directed graph shown, how many different paths of length 3 exist?



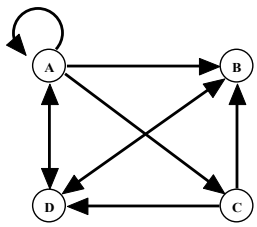
12. **05-06 C3 Graph Theory**

Given the directed graph below, how many different cycles can start at vertex B?



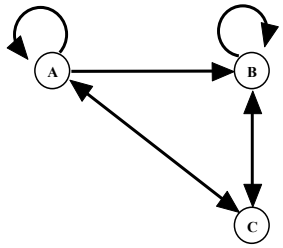
13. **06-07 C3 Graph Theory**

How many different cycles exist in the following directed graph?



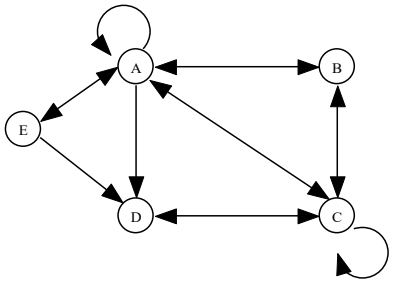
14. **06-07 C3 Graph Theory**

How many paths of length 3 exist in the following directed graph?



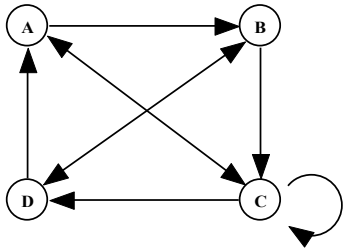
15. **07-08 C3 Graph Theory**

How many different cycles exist starting from node A in the following directed graph?



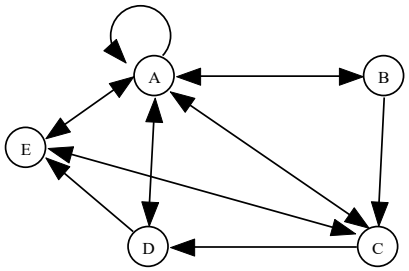
16. **07-08 C3 Graph Theory**

How many paths of length 3 exist in the following directed graph?



17. **08-09 C3 Graph Theory**

How many different cycles exist starting from node A in the following directed graph?

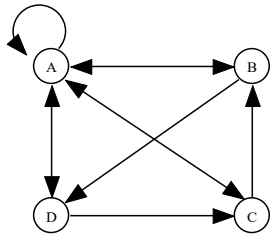


18. **08-09 C3 Graph Theory**

Given a directed graph having the following edges:
AB, BC, AC, AE, BD, EB, ED, CA, DE, CD
How many paths of length 2 exist?

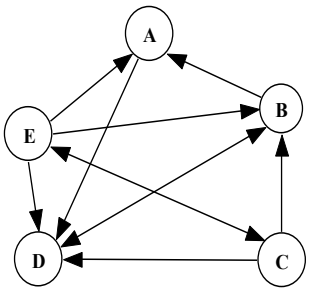
19. **09-10 C3 Graph Theory**

How many fewer cycles would there be from A if directed edge DA were removed?



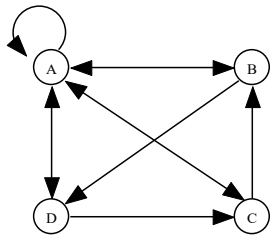
20. **09-10 C3 Graph Theory**

How many paths of length 2 exist in the following directed graph?



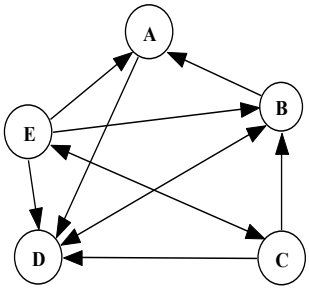
21. **10-11 C3 Graph Theory**

Which pair(s) of vertices has the most paths of length 3?



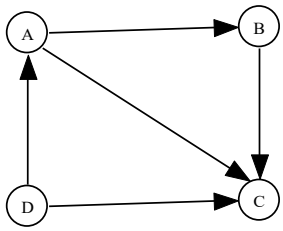
22. **10-11 C3 Graph Theory**

How many cycles are there from B in the following directed graph?



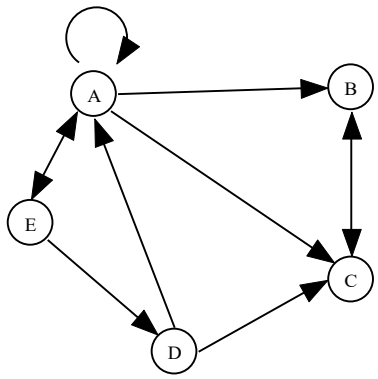
23. **11-12 C3 Graph Theory**

How many paths of length 4 exist in the directed graph at the right?



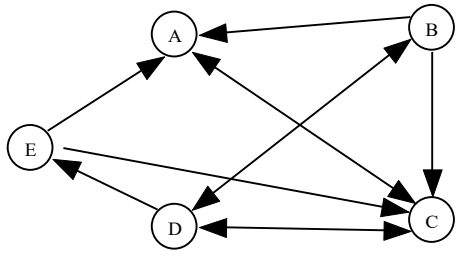
24. **11-12 C3 Graph Theory**

How many different cycles are there from A in the directed graph at the right?



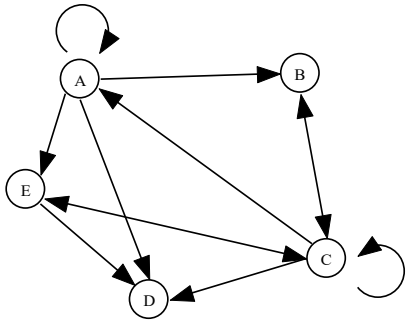
25. **12-13 C3 Graph Theory**

Between which pair(s) of vertices are there no paths of length 4 in the directed graph at the right?



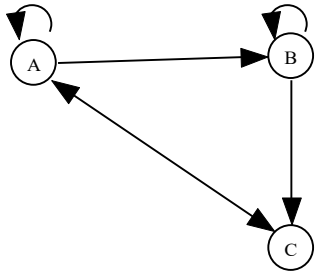
26. **12-13 C3 Graph Theory**

How many different cycles are there from C in the directed graph at the right?



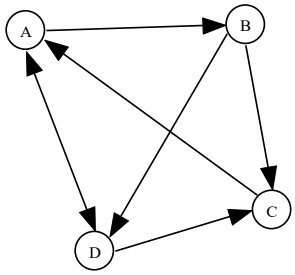
27. **13-14 C3 Graph Theory**

How many paths of length 3 are there in the directed graph?



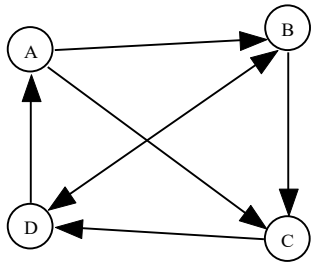
28. **13-14 C3 Graph Theory**

How many different cycles are there from A in the directed graph at the right?



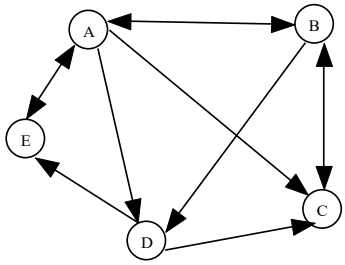
29. **14-15 C3 Graph Theory**

Between which pair(s) of vertices are there no paths of length 3 in the directed graph at the right?



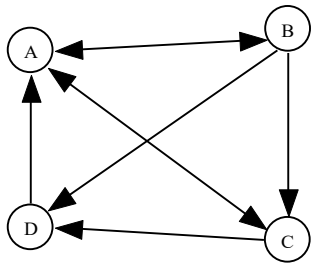
30. **14-15 C3 Graph Theory**

How many different cycles are there from B in the directed graph at the right?



31. **15-16 C3 Graph Theory**

How many paths of length 3 are there in the directed graph at the right?



32. **15-16 C3 Graph Theory**

How many different cycles are there from A in the directed graph at the right?

